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# Population genomics elucidates the trajectory of chicken domestication from Asia to Japan

Yuki Matsuda [a], Yuki Matsumoto [b,c], Yuki Higashiura [a], Andres Bendesky [d,e], Masaoki Tsudzuki [f], Tsuyoshi Shimmura [a,*]

[a] United Graduate School of Agricultural Science, Tokyo University of Agriculture and Technology, Saiwai-cho, Fuchu-shi, Tokyo 183-8509, Japan
[b] Research and Development Section, Anicom Specialty Medical Institute, Naka-ku, Chojamachi, Yokohama 231-0033, Japan
[c] Data Science Center, Azabu University, Sagamihara, Kanagawa 252-0206, Japan
[d] Zuckerman Mind Brain Behavior Institute, Columbia University, New York, 10027, USA
[e] Department of Ecology, Evolution and Environmental Biology, Columbia University, New York, 10027, USA
[f] Graduate School of Integrated Sciences for Life, Hiroshima University, Higashi-Hiroshima, Hiroshima 739-8528, Japan

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ABSTRACT

Chickens were domesticated over 4,000 years ago and are now the most popular poultry species in the world. In Japan, chickens are believed to have been first introduced around 2,000 years ago. Today, approximately 45 chicken breeds can be found in Japan, and they have unique and diverse phenotypic characteristics. However, their origins and histories have largely remained unexplored. Here, we newly sequenced whole genomes of 41 populations including 21 Japanese native chickens. By integrating them with published genomes to compile a total of 133 geographically comprehensive samples, we conducted population genomic analyses to elucidate the trajectory of their domestication from Asia to Japan. Our findings indicate that Japanese native chickens were derived from Southeast and East Asia, the centers of chicken domestication, through several separate events. The whole-genome phylogenetic analyses classified Japanese chickens into several groups and clarified the geographic origins of each group. These results provide new insights into the population structure, admixture history of Asian chicken breeds, and genomic trajectory of chickens from Asia to Japan.

## Introduction

Chickens (*Gallus gallus domesticus*) are the most popular poultry in the world. While they are recently bred for food production (Crawford, 1990), they were anciently used for ornamental, cultural, and recreational purposes (Bryant, 1991; Tsudzuki, 2003; Tixier-Boichard et al., 2011; Potts, 2012; Manyelo et al., 2020). The ancestor of domestic chickens is the red jungle fowl (RJF, *G. gallus*), and domestication has been estimated to have occurred approximately 4,000 to 10,000 years ago in South, Southeast, or East Asia (Tixier-Boichard et al., 2011; Ulfah et al., 2016; Lawal et al., 2020; Wang et al., 2020). Domesticated chickens then rapidly spread across the globe through multiple routes around 1,500 to 8,000 years ago (West and Zhou, 1988; Storey et al., 2012; Peters et al., 2022), and nowadays more than 1,600 chicken breeds have been observed worldwide (Scherf et al., 2015).

Archeological studies suggest that chickens were first introduced into Japan through the Korean Peninsula around 1,700 to 2,300 years ago during the Yayoi period (West and Zhou, 1988; Eda et al., 2023). Currently, approximately 45 chicken breeds are recognized in Japan, exhibiting unique and diverse phenotypic characteristics (Tsudzuki, 2003) (Fig. 1**a**). Based on morphological observations and literature review, Oana (Oana, 1951) hypothesized the origin and lineage of Japanese native chicken breeds. In this book, Japanese chicken breeds were subdivided into three groups: Jidori group, Shokoku group, and Shamo group. Jidori is a group of Japanese old-type chickens believed to have been introduced to Japan more than 2,000 years ago (Tsudzuki, 2003). Shokoku was introduced to Japan between 832 and 1,230 years ago during the Heian period and is thought to have contributed to the establishment of some Japanese breeds (Tadano et al., 2007). Shamo is a group of breeds for cockfighting and exhibits highly aggressive behavior (Komiyama et al., 2003; Bendesky et al., 2024). Other Japanese breeds were sporadically introduced into Japan or have been established

* Corresponding author at: United Graduate School of Agricultural Science, Tokyo University of Agriculture and Technology, Saiwai-cho, Fuchu-shi, Tokyo 183-8509, Japan.
*E-mail address:* shimmura@go.tuat.ac.jp (T. Shimmura).

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through crossbreeding with the breeds mentioned above (see materials and methods for details). These Japanese chickens are certified as indigenous breeds under the Japanese Agricultural Standards (https://www.maff.go.jp/e/policies/standard/jas, accessed on 19 Dec. 2024). Notably, 17 of these breeds are designated National Natural Treasures (Kuroda et al., 1987), ensuring their preservation as pure bloodlines. Therefore, Japanese native chickens represent invaluable bio-resources for studying the origin and trajectory of domestication in Asia. Additionally, the unique characteristics of Japanese chickens provide a valuable model for investigating how natural and artificial selection drive phenotypic evolution during animal domestication.

Previous analyses using mitochondrial DNA (Komiyama et al., 2003, 2004; Oka et al., 2007; Hata et al., 2020) and microsatellite DNA markers (Osman et al., 2006; Tadano et al., 2007, 2008, 2009) have reported the genetic characteristics of Japanese native chickens. Japanese chickens have a similar haplogroup composition with East Asian chickens and share a part of haplogroups with the Southeast Asian population (Hata et al., 2020). Among the Japanese population, the mitochondrial haplotype is roughly divided into two groups: the Shamo (gamecock) group and other breeds (Oka et al., 2007). The microsatellite DNA markers detected distinctive genetic characteristics in Japanese chicken breeds, which genetically differentiates them from chickens outside of Japan (Osman et al., 2006; Hata et al., 2020). However, genetic information acquired from mitochondrial DNA and microsatellites is highly limited, restricting the application of population structure and admixture analyses. Therefore, the domestication trajectory of Japanese native chickens remains largely uncovered. A recent RAD sequencing analysis (Velasco et al., 2024) reported the relationship among some Japanese chicken breeds, indicating that genome-wide SNPs provide a more precise assessment of population diversity and structure (Fischer et al., 2017; Zimmerman et al., 2020). Thus, whole-genome sequence comparisons are essential for better elucidating the origin and the history of the divergence of Japanese chickens.

Here, we performed the whole-genome sequencing and utilized a total of 41 populations including 21 Japanese native chicken breeds. By comparing these genomes with published genomes of indigenous breeds from China, Korea, Southeast Asia, and other regions, we showed the origin and trajectory of domestication of the chickens from Southeast and East Asia to Japan.

## Materials and methods

### *Ethic statement*

Animals were treated in accordance with the guidelines of Hiroshima University, Japan. All experimental protocols were approved by the Animal Experiment Committee of the university.

### *Sample collection and genome sequencing*

We sampled 41 populations from 31 breeds including 21 Japanese native breeds and 10 commercial breeds or breeds from other regions (Fig. 1**b**, Table 1, Table 2). Together with 8 populations sequenced in the previous study (Bendesky et al., 2024), we totally used 36 populations from 25 breeds of Japanese chickens in this study (Table 1). Among these samples, Gifu-jidori (GJD), Mie-jidori (MIE), and Tosa-jidori (TSA) belong to the Jidori group. Shokoku (SHK), Kurokashiwa (KKW), Minohiki (MHK), Totenko (TOT), and Onagadori (ONG) are thought to be the Shokoku group. Oshamo (OSM), Koshamo (KSM), Nankinshamo (NSM), Yamatogunkei (YSM), and Yakido (YKD) belong to the Shamo

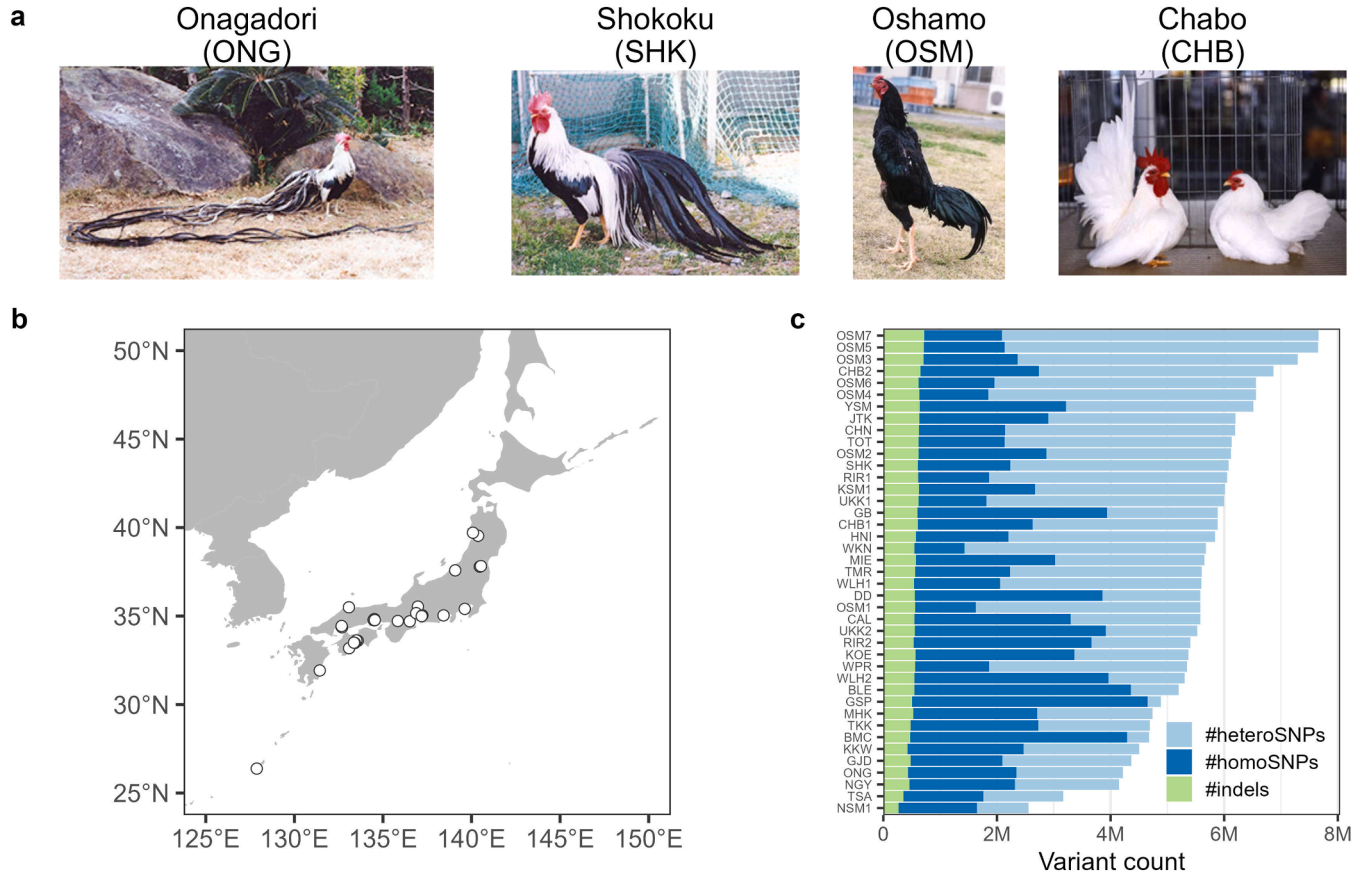



**Figure 1. Sample distribution and variant count (a)** Photos of representative Japanese chickens. Onagadori, Shokoku, Oshamo, and Chabo from left to right. **(b)** Map showing the distribution of samples newly sequenced in this study. **(c)** Bar chart representing the variant count of each sample. Green, blue, and light blue represent homozygous alleles, heterozygous alleles, and indels, respectively.

**Table 1**
Sample information of Japanese chickens used in this study.

| Sample name | Population | Breed | Number of individuals | Source |
|---|---|---|---|---|
| CHB1 | Chabo | Chabo | 16 | This study |
| CHB2 | Chabo-ABRC | Chabo | 10 | This study |
| CHN | Chan | Chan | 32 | This study |
| GJD | Gifujidori | Gifujidori | 5 | This study |
| HNI | Hinaidori | Hinaidori | 11 | This study |
| JTK | Jitokko | Jitokko | 2 | This study |
| KKW | Kurokashiwa | Kurokashiwa | 18 | This study |
| KOE | Koeyoshi | Koeyoshi | 11 | This study |
| KSM1 | koshamo-Kochi | koshamo | 11 | This study |
| MHK | Minohiki | Minohiki | 7 | This study |
| MIE | Miejidori | Miejidori | 13 | This study |
| NGY | Nagoya | Nagoya | 5 | This study |
| NSM1 | Nankinshamo-Fukushima | Nankinshamo | 2 | This study |
| ONG | Onagadori | Onagadori | 5 | This study |
| OSM1 | Oshamo-Fukushima | Oshamo | 16 | This study |
| OSM2 | Oshamo-Hyogo831 | Oshamo | 13 | This study |
| OSM3 | Oshamo-Hyogo833 | Oshamo | 17 | This study |
| OSM4 | Oshamo-Hiku-Hiroshima | Oshamo | 36 | This study |
| OSM5 | Oshamo-Hiku-Kochi | Oshamo | 10 | This study |
| OSM6 | Oshamo-Uwa-Hiroshima | Oshamo | 27 | This study |
| OSM7 | Oshamo-Uwa-Kochi | Oshamo | 16 | This study |
| SHK | Shokoku | Shokoku | 13 | This study |
| TKK | Tosakukin | Tosakukin | 6 | This study |
| TMR | Tomaru | Tomaru | 8 | This study |
| TOT | Totenko | Totenko | 20 | This study |
| TSA | Tosajidori | Tosajidori | 7 | This study |
| UKK1 | Ukokkei | Ukokkei | 6 | This study |
| UKK2 | Ukokkei-ABRC | Ukokkei | 10 | This study |
| YSM | Yamatogunkei | Yamatogunkei | 4 | This study |
| KMT | Kumamoto | Kumamoto | 1 | (Bendesky et al., 2024) |
| KSM2 | Koshamo-Fukushima | Koshamo | 1 | (Bendesky et al., 2024) |
| NSM2 | Nankinshamo-Hiroshima | Nankinshamo | 1 | (Bendesky et al., 2024) |
| STM | Satsumadori | Satsumadori | 1 | (Bendesky et al., 2024) |
| TSM | Tauchi | Tauchi | 1 | (Bendesky et al., 2024) |
| YKD1 | Yakido-Fukushima | Yakido-Fukushima | 1 | (Bendesky et al., 2024) |
| YKD2 | Yakido-Hyogo | Yakido-Hyogo | 1 | (Bendesky et al., 2024) |
| YKD3 | Yakido-Mie | Yakido-Mie | 1 | (Bendesky et al., 2024) |

group. Chabo (CHB) and Chan (CHN) are thought to be independently introduced into Japan. Other breeds (Hinaidori (HNI), Jitokko (JTK), Koeyoshi (KOE), Tomaru (TMR), Nagoya (NGY), Tosakukin (TKK), Kumamoto (KMT), and Satsumadori (STM)) are thought to have been established by subsequent crossbreeding. For commercial breeds or breeds from other regions, we sampled 12 populations from 10 breeds to study the trajectory of chicken domestication outside Asia (Table 2). These samples include Blown-Leghorn (BLE), Black-Minorca (BMC), White-Wyandotte (CAL), Dandarlawi (DD), Game-Bantam (GB), Fayoumi-GSP (GSP), Rhode-Island-Red (RIR), White-Cornish (WKN), White-Leghorn (WLH), and White-Plymouth-Rock (WPR).

We obtained blood samples of Japanese chickens provided by breeders and Livestock Experiment Stations in Japan with informed consent (Table S1). Blood samples were collected from the wing vein with heparin as an anticoagulant and stored at -20°C until DNA extraction. To determine the representative variants of each population

**Table 2**
Sample information of other regions chickens sequenced in this study.

| Sample name | Population | Number of individuals | Origin | Source |
|---|---|---|---|---|
| BLE | Brown-Leghorn | 10 | Italy | This study |
| BMC | Black-Minorca | 10 | Spain | This study |
| CAL | White-Wyandotte | 10 | USA | This study |
| DD | Dandarlawi | 10 | Egypt | This study |
| GB | Game-Bantam | 10 | UK | This study |
| GSP | Fayoumi-GSP | 10 | Egypt | This study |
| RIR1 | Rhode-island-red | 6 | USA | This study |
| RIR2 | Rhode-island-red | 10 | USA | This study |
| WKN | White-cornish | 14 | UK | This study |
| WLH1 | White-Leghorn | 11 | Italy | This study |
| WLH2 | White-Leghorn | 10 | Italy | This study |
| WPR | White-Plymouth-Rock | 9 | USA | This study |

and evaluate population-level phylogeny, we conducted pooled sequencing. The blood samples were pooled per population. DNA was extracted using the DNeasy blood and tissue kit (QIAGEN) according to the standard protocol of the kit. The quantity and quality of the DNA samples were evaluated using a Qubit fluorometer and agarose gel electrophoresis. We then created Tn5-tagmented, 9-PCR cycles amplified, whole-genome sequencing libraries (Picelli et al., 2014) of an average insert size of 300 to 400 bp. Libraries from each DNA sample were sequenced with 150 bp paired-end reads using an Illumina Novaseq 6000 or HiSeq Ten X sequencer (Table S1).

Additionally, we downloaded the sequencing data of 99 samples from 4 NCBI BioProjects: 12 RJFs, 6 Thai breeds, 7 Vietnamese breeds, and 54 Chinese breeds from PRJEB42301 (Wang et al., 2020), 5 Korean breeds from PRJNA241461 (Kwak et al., 2014), 3 Chinese gamecocks from PRJNA627080 (Luo et al., 2020), 8 Japanese populations, 1 Chile breed, and 3 gamecocks in Southeast Asia from PRJNA973554 (Bendesky et al., 2024). We selected *G. g. spadiceus* for RJF because this subspecies is estimated to be the closest ancestor of domesticated chickens (Wang et al., 2020; Eda, 2021). Accession numbers for each sample are listed in Supplementary Table S3.

### *Read mapping and variant calling*

We checked the quality of raw reads using FastQC (https://www.bioinformatics.babraham.ac.uk/projects/fastqc) and trimmed adapters using trimmomatic 0.39 (Bolger et al., 2014) with "ILLUMINACLIP:NexteraPE-PE.fa:2:30:10 LEADING:20 TRAINING:20 SLIDINGWINDOW:4:20 MINLEN:30" parameters. For the downloaded data, the quality of the raw reads was checked and controlled using fastp 0.23.4 (Chen et al., 2018). The read pairs containing $\geq$ 30% of bases with base quality $Q \leq 30$ were removed by "-q 30 -u 30" parameters. We used Burrows-Wheeler Aligner (BWA) version 0.7.17 to map the clean sequencing reads to the Gallus gallus reference genome (bGalGal1.mat.broiler.GRCg7b, https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_016699485.2/). The BWA-MEM algorithm (Li, 2013) was used to generate sam file for each sample. SAMtools version 1.18 (Li et al., 2009) was used to sort reads, mark duplicated reads in sam files, and then to convert sam files to bam files. Variants were called per population for newly sequenced samples in this study and per individual for previously published samples. The bam files were used to muti-way pileup and variant calling using the mpileup command with the

options "–max-depth 1000 –no-BAQ" and call command with "-vm" option (Danecek et al., 2014) implemented in BCFtools version 1.18 (Li, 2011). To mitigate the effects of sequencing errors caused by pooled sequencing, we subsampled the variants for each population to obtain uniform coverage along with the protocol described in a previous study (Kurland et al., 2019). In short, sites with QUAL>30 and sequence coverage from × 1/2 to × 2 of the mode of coverage were subsampled using the filter command in BCFtools.

*Filtering samples and SNPs*

We excluded 6 downloaded samples (ERR5036133, ERR5036134, ERR5036135, ERR5036136, ERR5036137, and ERR5036705) because of low genome coverage (average coverage < 90%, Table S3). ERR5036138 was also excluded because all other samples in the same population were removed by the above filtering. The samples used in this study were sequenced by two different instruments: Illumina HiSeq and NovaSeq. The population structure was affected by the difference of the sequencer (Fig. S1a). To remove this batch effect, we performed GWAS analysis using the sequence instruments as phenotype and removed SNPs with highly significant association. The raw p-values were adjusted using the Benjamini-Hochberg (BH) multiple testing correction method (Benjamini and Hochberg, 1995). SNPs with p-values lower than 0.05 were removed (Fig. S1b), and the subsampled variants were used for the following analysis.

*Haplotype analysis of mitochondrial whole genome*

BCFtools was used for variant calling of mitochondrial whole genome. Reads mapped on mitochondria genome were piled up using "–region NC_053523.1", and then all variants were called. VCF files were converted to FASTA format using the vcf2phylip software (Ortiz et al., 2019). RapidNJ (Simonsen et al., 2008) was used to infer the neighbor-joining tree. Each sample was classified into mitochondrial haplogroup by MitoToolPy software (Peng et al., 2015) using "-s chicken -r whole" arguments.

*Population structure analysis*

For the population structure analysis, we subsampled the biallelic SNPs using the BCFtools view command with the arguments "-v snps -m2 -M2". To avoid the effect of rare variants, we removed sites with minor allele frequencies less than 0.01 using PLINK v2.00 alpha (Chang et al., 2015). Then, we pruned linkage disequilibrium of variants with the parameter "–indep-pairwise 150 50 0.2". The pruned variants were used for principal component analysis using PLINK. We ran ADMIXTURE v1.3 (Alexander et al., 2009) at K = 1 to 15 using the same variant set and calculated the cross-validation error by default parameters. We used RapidNJ to infer the neighbor-joining tree using the subsampled variants.

*TreeMix analysis*

We used TreeMix v1.13 (Pickrell and Pritchard, 2012) to estimate the historical relationships and admixture events between Japanese chicken population and other chicken breeds. For the TreeMix analysis, we excluded SNPs on sex chromosomes and mitochondria from pruned variants using PLINK with "–not-chr NC_052571.1, NC_052572.1, NC_053523.1" arguments. We ran TreeMix with 1 to 8 migration edges and 5 iterations for each migration edges. The optimal number of migration edges was selected based on the rate of change in log likelihood using the OptM R package (Fitak, 2021).

*Visualization*

We used R version 4.4.1 (R Core Team, 2024) with the packages tidyverse (Wickham et al., 2019), phytools (Revell, 2024), and ggtree (Yu et al., 2017) for data manipulation and visualization.

## Results

*Genome sequence and read mapping*

In this study, we newly sampled a total of 41 populations from 31 breeds which include 21 Japanese native breeds and 10 commercial or other regions breeds (Fig. 1**b**, Table 1, Table 2). After pooling the blood samples of 2 to 36 individuals in each population, we sequenced these samples to sequence depth ranging from 7.07 × to 21.2 × and genome coverages >94% (Table S2). To obtain representative genotypes for each population, we first filtered out rare variants and potential sequencing errors by site quality and sequence depth. Sites with low quality or extremely high/low depth were filtered out, and we identified 267,128 to 719,026 indels, 880,800 to 4,148,933 homogenous SNP sites, and 230,519 to 5,570,117 heterogenous SNP sites (Fig. 1**c**). In total, we identified 2,555,856 to 7,658,207 variants for each population.

We downloaded additional sequencing data of 99 samples from 4 NCBI BioProjects. These samples included 12 RJFs, 6 Thai samples, 7 Vietnamese samples, 5 Korean samples, 6 gamecocks, 54 Chinese samples, 8 Japanese samples, and 1 Chile sample. After incorporating these public sequence data, we obtained 29,310,883 variants, including 26,023,703 SNPs. Filtering by linkage disequilibrium and minor allele frequency subsampled 4,622,867 SNPs.

These samples contained the sequences from illumina HiSeq and NovaSeq, but the differences of sequencers affected the genetic structure of the samples (Fig. S1a). To remove this effect, we ran GWAS using the sequencers as phenotypes and filtered out SNPs with significant associations (adjusted p-value<0.05; 319,058 SNPs; Fig. S1b). We confirmed that the first principal component was not affected by the sequencers (**Figure S1c**) and used the filtered 4,303,809 SNPs in the subsequent analyses. Although the batch effect may remain in the lower components, we determined that it would be difficult to further distinguish the minor batch effect from the true population structure.

*Maternal origin of Japanese chicken breeds*

To elucidate the maternal origins of Japanese chickens, we conducted the haplogroup classification using mitochondrial whole-genome sequence. All variants in the mitochondrial genome were used to infer the neighbor-joining tree and haplogroup classification (Fig. 2). The chicken populations used in this study were classified into 7 main haplogroups (A to G) and 36 sub-haplogroups (Fig. 2**a**), which was supported by the neighbor-joining tree of the mitochondrial whole genome.

Japanese chicken populations were classified into 16 sub-haplogroups (Fig. 2**b**). Although some populations possess multiple haplogroups, Japanese chickens could be separated into 5 main haplogroups: A, B, C/D, E, and intermediate groups. This suggests that Japanese chickens were derived from several haplogroups. The neighbor-joining tree of the mitochondrial genome also suggests that Japanese chickens were present in multiple lineages (Fig. 2**a**). A and C haplotypes were observed in populations of the northern part of China (Shanxi, Beijing, and Shandong) and a part of Korean native chickens (KNC). The B haplogroups were shared with populations from southern to central China (Jiangsu, Jiangxi, and Guangxi) and Southeast Asia (Thailand and Vietnam). Japanese chickens possessing D haplotype variants were derived from populations in Southeast Asia. These results indicate that Japanese chicken breeds have multiple maternal origins in Southeast and East Asia.

*Population structure and admixture of chicken breeds in Asia*

To reveal the population structure of chicken breeds in Southeast and

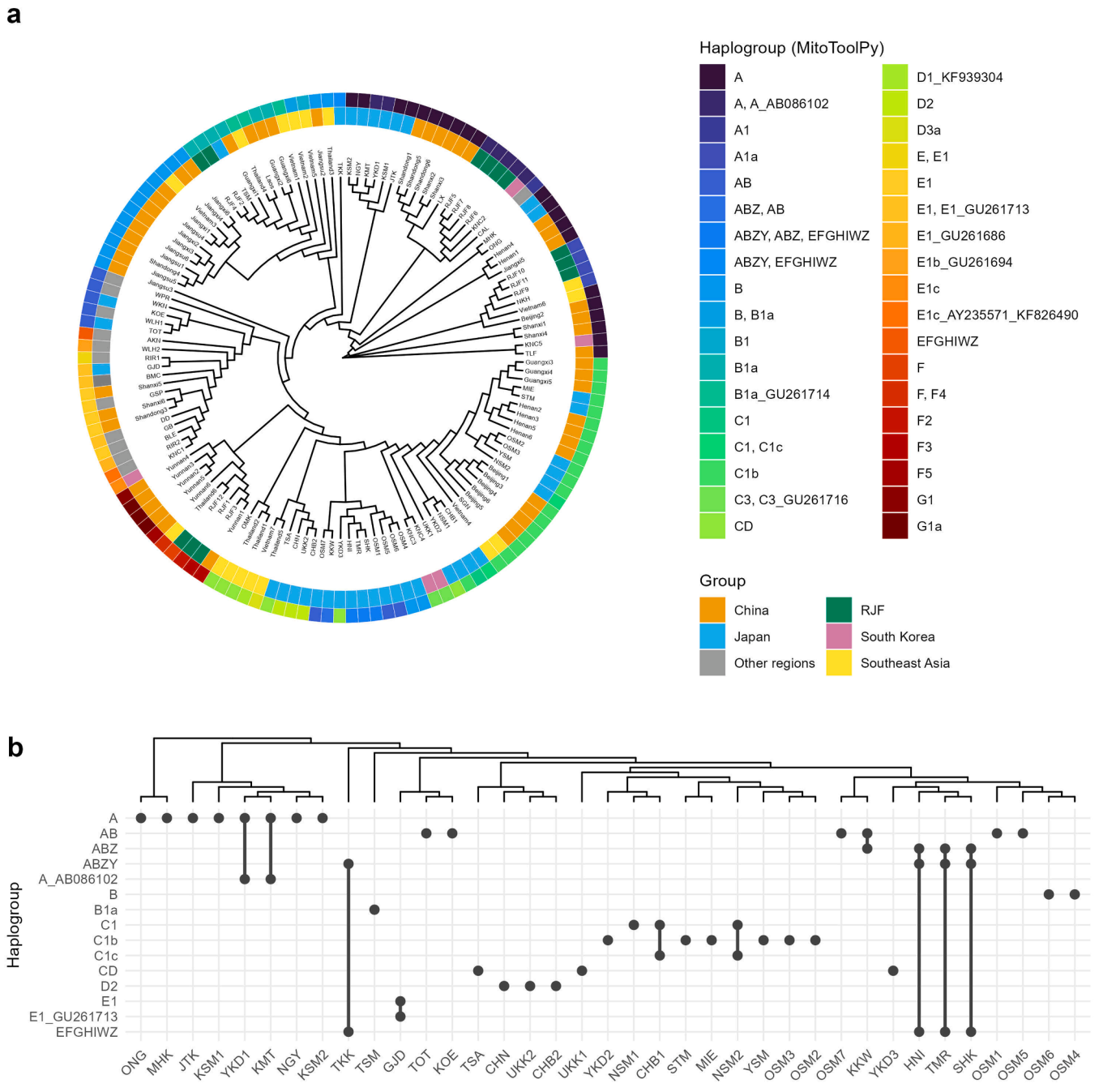



**Figure 2. Phylogenetic tree of mitochondrial genome and inferred haplogroups of Japanese chickens (a)** Neighbor-joining phylogenetic tree of mitochondrial whole-genome. Heatmap represents the haplogroups classified by MitoToolPy. **(b)** Occurrence of mitochondrial haplotypes in Japanese chickens. Neighbor-joining phylogenetic tree of mitochondrial whole-genome are shown for a subset of Japanese breeds.

East Asia, we conducted PCA analysis using LD-pruned SNP dataset of whole genome. While PC1 and PC2 roughly separated the chicken populations into Asian breeds and breeds from other regions, PC3 and PC4 primarily distinguished chicken populations by their geographical origins (Fig. 3**a, b**). Subsequently, we ran ADMIXTURE analysis to estimate the ancestries of these populations. The cross-validation error suggests that chicken populations were most likely to have K=1 ancestry, indicating that the chicken populations have no clearly different structure (Fig. 3**c**). Although the most possible structure has no multiple ancestry states, we show the results of ADMIXTURE analysis of K=2 to K=4 to evaluate the admixture states of these chicken breeds (Fig. 3**d**). In any ancestry states, the Japanese population has a relatively high degree of admixture.

To infer the origin of Japanese chicken breeds, we reconstructed the neighbor-joining tree based on whole genomic SNPs (Fig. 4**a**). The neighbor-joining tree separated Japanese breeds into several clades (see nodes 1 to 5 in Fig. 4**a**). Most Japanese gamecocks (Shamo) clustered into the monophyletic clade, indicating the genetic closeness of these breeds (node 1). Chan (CHN), Koeyoshi (KOE), Jitokko (JTK), Miejidori (MIE), and Chabo (CHB) were clustered in a clade (node 2). Shokoku (SHK), Kurokashiwa (KKW), Onagadori (ONG), Totenko (Totenko), and Tomaru (Tomaru) formed a monophyletic clade (node 3). Breeds that were considered as the Shokoku group were classified in this clade except for Minohiki (MHK). Some Japanese gamecocks, such as Yakido (YKD) and Nankinshamo (NSM1), were included in node 4 and this node also contained the gamecocks of Southeast Asia. Three Japanese

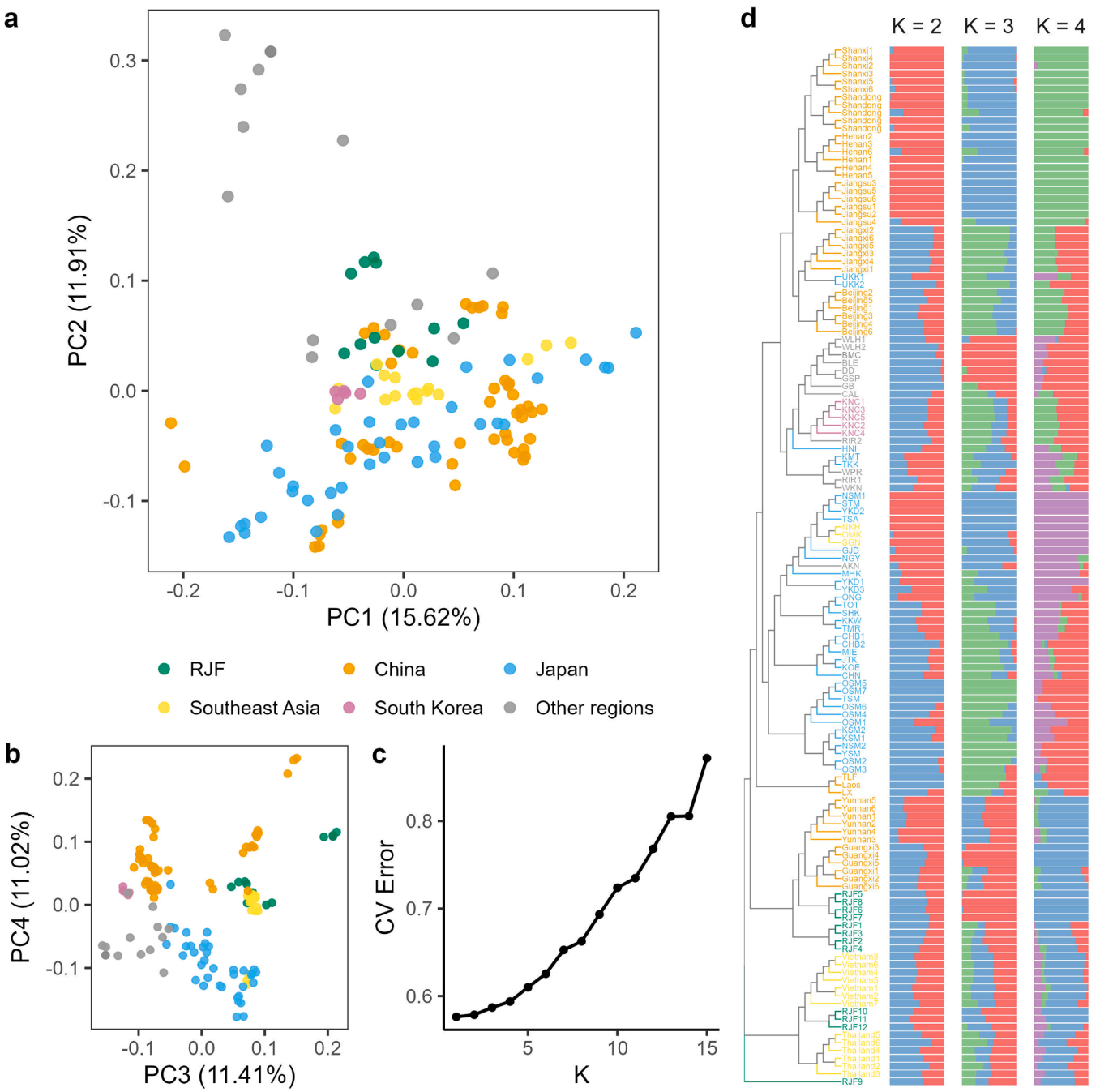



**Figure 3. Population structure of RJF and chicken in Asia (a)** Plot for PC1 and PC2 of genetic variation in RJF and chicken in Asia. **(b)** Plot for PC3 and PC4 of genetic variation in RJF and chicken in Asia. **(c)** Cross-Validation error of each K of ADMIXTURE analysis. The lowest cross-validation error was observed in K=1. **(d)** Neighbor-joining phylogenetic tree and the results of ADMIXTURE analysis at K=2 to K=4.

breeds—Hinaidori (HNI), Kumamoto (KMT), and Tosakukin (TKK)—were clustered with the other regions' breeds (node 5). In contrast to the other Japanese breeds, Ukokkei (UKK) was genetically close to the Chinese Jiangxi population.

To better understand the admixture history and phylogenetic relationship within the chicken populations, we reconstructed the maximum likelihood (ML) tree using TreeMix software and estimated the migration edges among populations (Fig. 4**b–d**). Based on the neighbor-joining tree reconstructed using whole genomic SNPs, Japanese chicken breeds were clustered into 5 groups (Fig. 4**b**). In the TreeMix analysis with 1 to 8 migration edges, the second-order rate of change ($\Delta m$) by OptM indicated that the optimal number of migrations was 1 (Fig. 4**c**). Notably, Japanese chickens did not form a monophyletic group but were found in multiple lineages (Fig. 4**d**), indicating that there were several migrations of chickens into Japan. Japanese gamecocks clustered with Chinese gamecocks and two groups of Japanese breeds (Node 2 and Node 3) were closely related to population of Beijing, South Korea, and Jiangxi. Japanese group of node 4, which included some gamecocks, was clustered with the gamecocks of Southeast Asia. Furthermore, one migration edge was estimated from the ancestral node of Japanese group of node 4 and gamecocks of Southeast Asia to Shokoku group.

## Discussion

In this study, we newly sequenced the whole genomes of 41 populations including 21 Japanese breeds and integrated them with

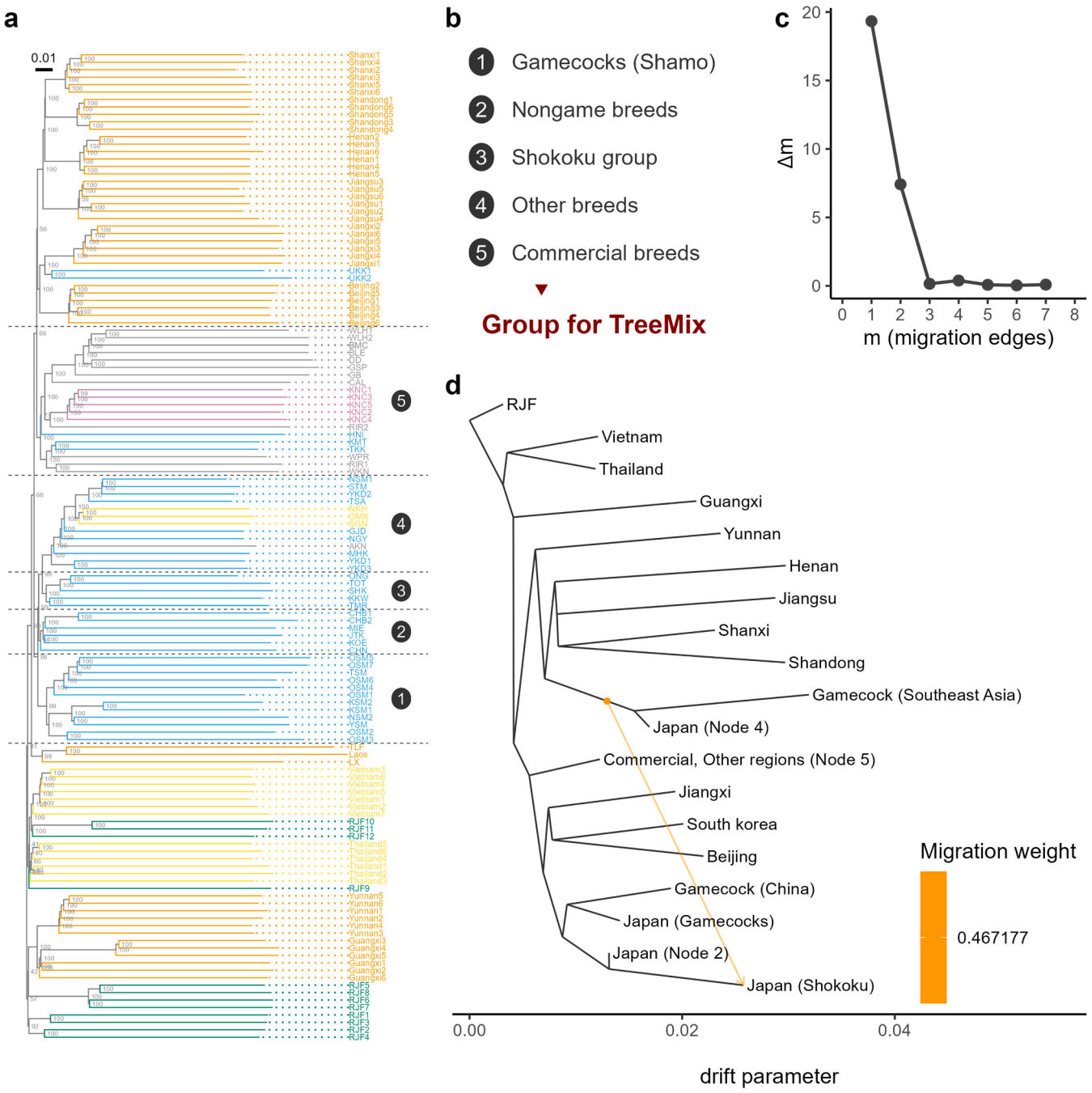



**Figure 4. TreeMix analysis of Japanese and other chicken populations (a)** Neighbor-joining tree of Japanese chickens inferred from whole genomic SNPs. The numbered nodes (1–5) are the clades including Japanese chickens and are mentioned in the main text. **(b)** Groups of Japanese chickens for TreeMix analysis. The numbers correspond to the neighbor-joining tree. **(c)** The second-order rate of change ($\Delta m$) across each migration edges. The highest $\Delta m$ is observed in 1 migration edge. **(d)** Maximum likelihood tree of chicken populations inferred using TreeMix. The 1 migration edges are shown.

published genomes of chickens from Southeast and East Asia. We leveraged a total of 133 geographically comprehensive samples to elucidate the domestication trajectory of these populations. To our knowledge, this work is the first report on the whole-genome population genomics of Japanese chickens by comparing with Southeast and East Asia as the centers of chicken domestication.

To elucidate the whole-genome population structure of chicken breeds in Southeast and East Asia, including Japan, we conducted a series of population genomic analyses. The principal component analysis and phylogenetic analysis revealed that the chicken breeds of Southeast Asia and Southern China are closely related to the red jungle fowls (RJFs), which supports the hypothesis that domestic chickens originated in these regions (Wang et al., 2020). However, while most RJFs clustered as an outgroup of domestic chickens, some RJFs were nested in the Thai and Vietnamese populations. One possibility is that domestic chickens have multiple origins in these regions and were repeatedly domesticated from wild jungle fowls (Liu et al., 2006a). Alternatively, the genetic introgression might be occurred from domestic indigenous chickens into RJFs where these populations co-exist (Berthouly et al., 2009; Wang et al., 2020; Wu et al., 2020, 2023). Further studies are needed in the near future to fully understand the early domestication of chickens by leveraging geographically and temporally comprehensive samples, including ancient chicken DNA sequences (Girdland Flink et al., 2014; Wu et al., 2023).

The morphological observations and extensive literature review hypothesized that Japanese chickens are classified into three main groups: Jidori group (traditional Japanese breeds), Shokoku group, and Shamo group (Gamecocks) (Oana, 1951). Although the phylogenetic relationships of Japanese native chickens have been studied using mitochondrial D-loop sequences and microsatellite DNA markers (Komiyama et al., 2003, 2004; Osman et al., 2006; Oka et al., 2007; Tadano et al., 2007; Hata et al., 2020), the genetic relationships of Japanese chickens did not align with the literature study (Oka et al., 2007; Hata et al., 2020). Consistent with these previous studies, our whole mitochondrial genome analysis indicated that Japanese chickens were derived from multiple clades, mainly from Chinese chicken populations, suggesting that Japanese chickens have multiple maternal origins. However, mitochondrial genome, which is inherited maternally and non-recombining, and microsatellites provide limited information to reveal the genome-wide population structure and admixture (Zimmerman et al., 2020). Presumably reflecting this, any genetic studies using mitochondrial DNA and microsatellite markers are not well fitted with the literature study (Oana, 1951).

Our whole-genome analysis resolved the discrepancies between the literature study and the phylogenetic analysis. In contrast to the mitochondrial DNA, the neighbor-joining tree based on the whole genome divided Japanese chickens into several groups (Fig. 4**a**). Interestingly, most Japanese gamecocks (Shamo) and the Shokoku group each formed distinct monophyletic clades (Node 1 and 3, respectively). Kurokashiwa (KKW), Totenko (TOT), and Onagadori (ONG) are thought to be established based on Shokoku (SHK) by their morphological characteristics (Oana, 1951; Tadano et al., 2007). Our result confirmed the genetic closeness of these breeds, which is consistent with previous hypotheses. Although some breeds, such as Yakido (YKD) among gamecocks and Minohiki (MHK) among the Shokoku group, were found in other clades. This suggests that the genetic introgression occurred through cross-breeding with other breeds. The relatively high degree of admixture in the Japanese chicken population also suggests the introgression among Japanese chickens. Furthermore, three Japanese breeds—Hinaidori (HNI), Kumamoto (KMT), and Tosakukin (TKK)—clustered with commercial breeds or breeds from other regions (Node 5). These breeds have been established for meat and egg production (Tsudzuki, 2003). The phylogenetic analysis shows that commercial breeds from other regions contributed to the development of these Japanese breeds.

Moreover, the reconstructed phylogenetic tree showed the genetic relationships between each group and chicken breeds in Southeast and East Asia (Fig. 4**d**), suggesting their geographic origins. The Shokoku group and another group of Japanese chickens (Node 2) appear to have originated from populations in Beijing, South Korea, and Jiangxi. This result may reflect the historical introduction of these Japanese breeds from China through the Korean peninsula (West and Zhou, 1988; Eda et al., 2023). Most Japanese gamecocks clustered with Chinese gamecocks. Although previous studies suggested multiple origins of Japanese gamecocks from China and Southeast Asia (Komiyama et al., 2003; Liu et al., 2006b), our phylogenetic analysis indicated that the direct ancestors of Japanese gamecocks were Chinese gamecocks. In contrast, another group of Japanese breeds (Node 4) was closely related to gamecocks from Southeast Asia and other parts of China, suggesting the geographic origin of this group differs from other Japanese breeds. Since this group included some gamecocks, it is possible that Japanese gamecocks were established from not only Chinese gamecocks but also gamecocks from Southeast Asia. Additionally, one migration edge connecting branches of Japanese breeds was estimated, which may indicate that the Japanese chicken population was admixed after their introduction into Japan. Taken together, the whole-genome phylogeny classified Japanese chickens into several groups, consistent with classification proposed in previous literature study (Oana, 1951). These analyses indicated that each group was introduced from distinct geographic origins (Fig. S2). Japanese native chickens were primarily introduced from northern part of China, probably through the Korean peninsula, and most Japanese gamecocks were derived from Chinese gamecocks. However, at least a part of Japanese breeds has been influenced by the introgression of chicken breeds in Southeast Asia. Some commercial Japanese breeds, such as Hinaidori (HNI), Kumamoto (KMT), and Tosakukin (TKK), were introduced or established from the breeds of other regions.

In this study, we unveiled the phylogenetic relationships of Japanese native chickens and their origins in Southeast and East Asia. These findings will provide significant insights into the preservation and breed improvement for the future. For instance, the genetic distance between populations used for cross-breeding influences the traits of the breeds and may lead to inbreeding depression (Lynch, 1991; Leroy, 2014). In conclusion, our analysis unveiled the population structure of indigenous chickens in Southeast and East Asia and revealed that chickens were introduced into Japan multiple times to construct the admixed population of Japanese native chickens. These findings shed light on the population structure and admixture history of these chicken breeds and emphasize the importance of whole genome sequences in reconstructing population trajectories.

## Author contributions

T.S. supervised the research. Y.Matsuda, T.S., and M.T. designed the research. M.T. and T.S. collected samples. Y.H. performed DNA extraction. A.B. conducted genome sequencing. Y.Matsuda, Y.Matsumoto, and T.S. analysed data. Y.Matsuda and T.S. wrote the paper. All authors have read and agreed to the published version of the manuscript.

## Code Availability

Codes for this study are archived in the GitHub repository: https://github.com/ymat2/matsuda2025jnc.

## Declaration of competing interest

The authors declare no competing interests.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.psj.2025.105060.

## Data availability

All supplementary data for this study are available within the paper and its Supplementary Information files. The nucleotide sequences of the Japanese chicken breeds have been deposited in the DDBJ Sequence Read Archive under Bioproject PRJDB19943.

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